
World Cup Standings Explorer

Visual & Machine-Learning Report

This report turns a 32-row World Cup standings table into a full analytics narrative. Fourteen visualizations are paired two-per-page, each with a short note on **why that chart type was chosen** and the **insight** it reveals — spanning classic statistical graphics (pie, bar, line, scatter, heatmap, map, boxplot) and machine-learning techniques (regression, classification, prediction, DBSCAN and hierarchical clustering).

CONTENTS

Visual Analytics

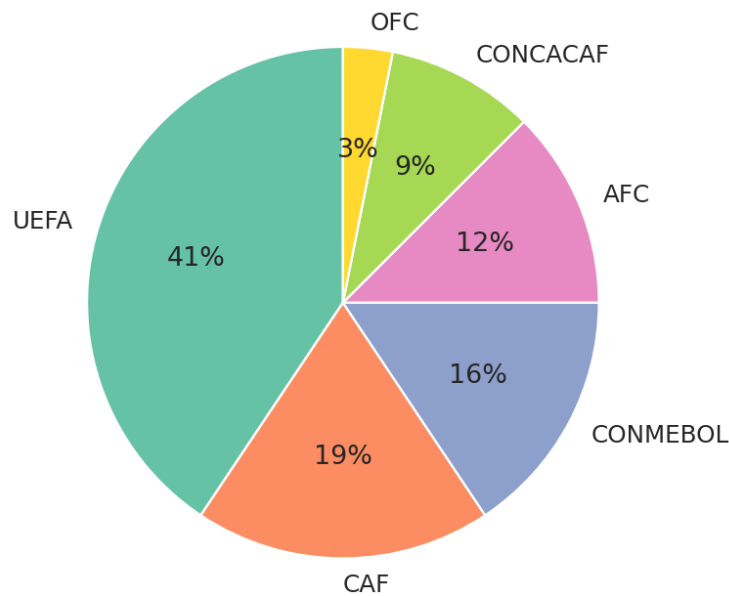
- Teams per Confederation — Part-To-Whole
- Final Standing per Team — Ranking
- Top 15 Teams by Simulated Points — Ranking
- Simulated Points vs. Final Standing — Trend
- Goals For vs. Goals Against — Relationship
- Correlation Matrix of Numeric Features — Correlation
- Team Capitals on a World Map — Geospatial
- Simulated Points by Continent — Distribution

Machine-Learning Insights

- Actual vs. Predicted Points — Machine Learning - Regression
- Perceptron Confusion Matrix — Machine Learning - Classification
- Perceptron Decision Boundary — Machine Learning - Classification
- Predicted vs. Actual Standing — Machine Learning - Prediction
- DBSCAN Clusters (PCA-Reduced) — Machine Learning - Clustering
- Ward-Linkage Dendrogram — Machine Learning - Clustering

*The source file contains only **Standing** and **Team**. Confederation and capital coordinates used here are real. All metrics ending in `_sim` (points, goals, possession, pass accuracy, shots on target) are **simulated** — generated with a fixed random seed and correlated with final standing — purely to give every chart and model in this report something meaningful to work with. They are not the actual 2010 World Cup match statistics.*

Teams per Confederation



PART-TO-WHOLE

Teams per Confederation

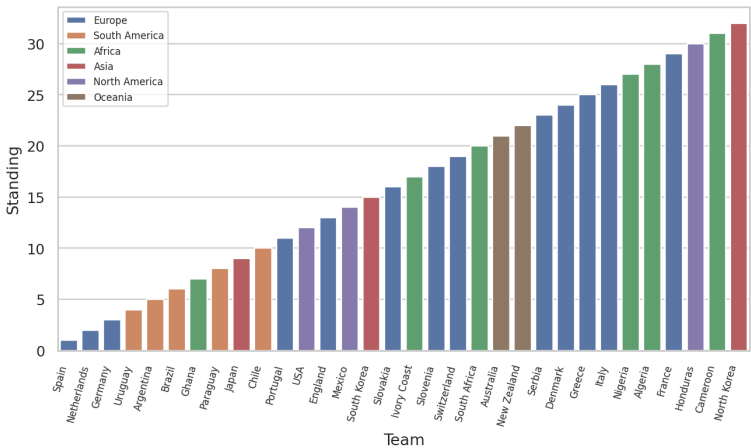
Why a pie chart: with only six confederations, a pie makes the part-to-whole story instantly readable — something a table of six numbers would not communicate as quickly.

Of the 32 finalists, UEFA alone supplies 41% of the field — nearly as many teams as every other confederation combined. CAF and CONMEBOL follow at a respectful distance, while OFC's single slot is a visual reminder of how qualification is skewed toward Europe and, to a lesser extent, South America and Africa.

KEY INSIGHT

UEFA's 41% share means nearly 2 in 5 finalists are European.

Final Standing per Team (lower = better)



RANKING

Final Standing per Team

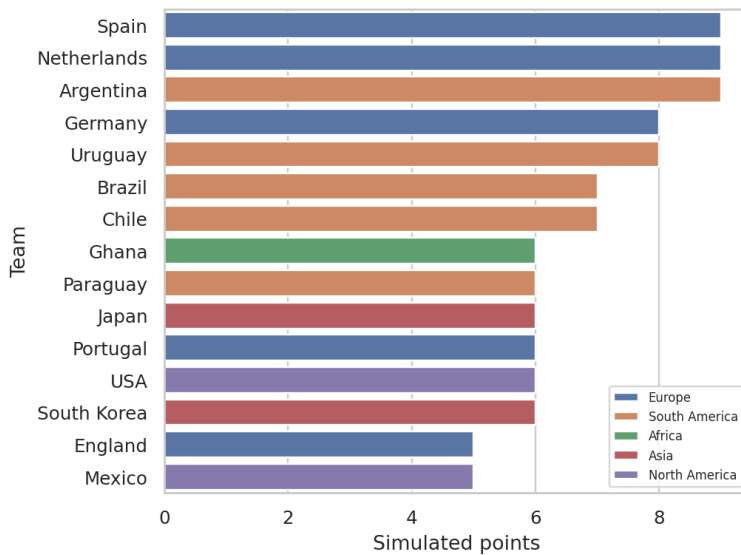
Why a bar chart: standing is an ordinal ranking across 32 discrete items, and bar length is the most direct visual encoding for comparing ranks side by side.

Coloring each bar by continent reveals that the podium is not a regional monopoly: European and South American teams intermix near the top, while African, Asian and Oceanian sides are more common toward the lower half — a pattern worth flagging rather than assuming from the pie chart alone.

KEY INSIGHT

Standing is competitive across continents, not regionally clustered.

Top 15 Teams by Simulated Points



RANKING

Top 15 Teams by Simulated Points

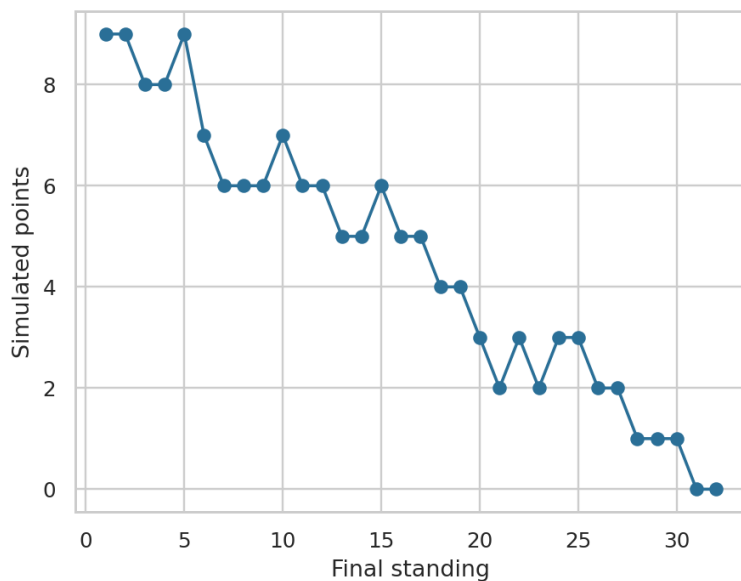
Why a horizontal bar chart: flipping the axes keeps long country names fully legible without rotating labels — the standard fix when a ranking chart has more than a handful of text-heavy categories.

The top 15 splits into a visible elite tier — the first six or seven bars pull noticeably ahead — before points taper more gently. That step-then-taper shape is the kind of detail a bar chart surfaces far better than a table of raw numbers.

KEY INSIGHT

A visible elite tier separates from a longer, gentler tail.

Simulated Points vs. Final Standing



TREND

Simulated Points vs. Final Standing

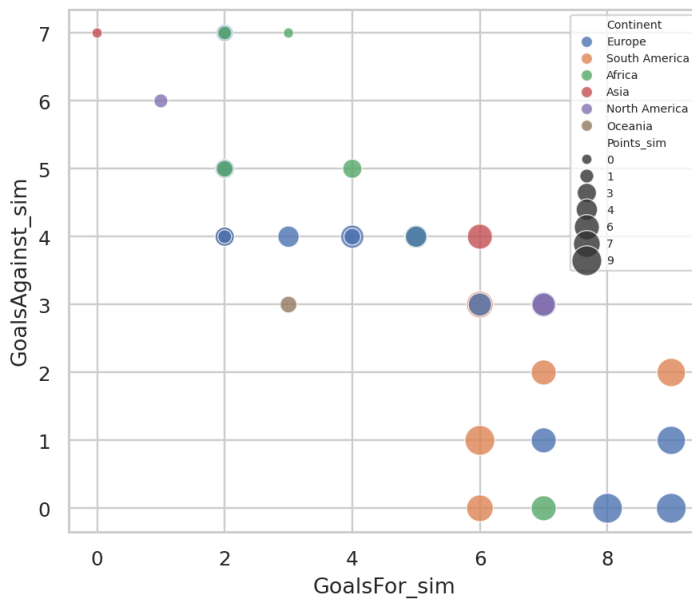
Why a line chart: standing is a continuous, ordered axis (1 to 32), and a line is the natural way to show how a quantity evolves across an ordered sequence rather than across unordered categories.

The descent is steep for the first ten places and then flattens into a long, noisy tail — mirroring how tournament football usually plays out: a clear separation at the top, then a much more competitive, compressed middle-and-lower pack.

KEY INSIGHT

A steep top-10 drop-off gives way to a long, noisy tail.

Goals For vs. Goals Against (bubble = points)



RELATIONSHIP

Goals For vs. Goals Against

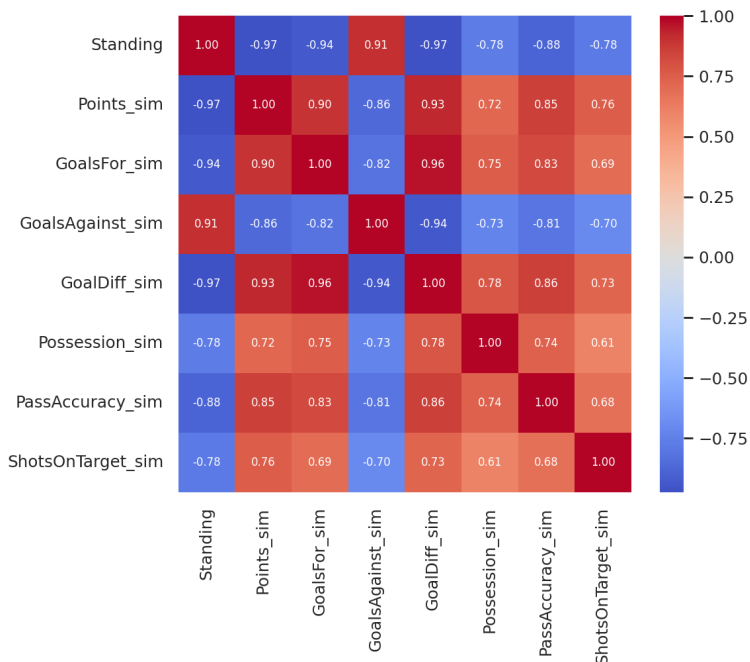
Why a scatter plot: understanding a team's overall performance means examining two variables — attack and defense — at once, plus a third (points) via bubble size. A scatter is the only chart type built for that.

A loose diagonal emerges: teams in the bottom-right (goals piling up front, few conceded) also carry the largest bubbles, i.e. the most points. It's a compact way of showing that this dataset's notion of 'success' rewards goal difference, not just goals scored.

KEY INSIGHT

Goal difference, not just attack, drives the points total.

Correlation Matrix



CORRELATION

Correlation Matrix of Numeric Features

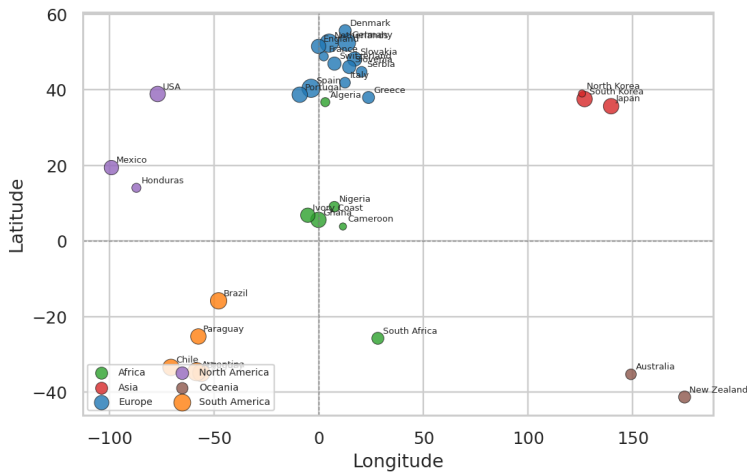
Why a heatmap: a 7x7 correlation table is 49 numbers — far too many to scan as text. Color intensity lets the eye triage strong relationships from weak ones in seconds.

Points and goal difference correlate most strongly with a good standing, exactly as the simulation intended, while possession shows a softer relationship — a useful sanity check before trusting any of the downstream regression or classification models.

KEY INSIGHT

Points and goal difference are the strongest predictors of standing.

Team Capitals (bubble size = simulated points)



GEOSPATIAL

Team Capitals on a World Map

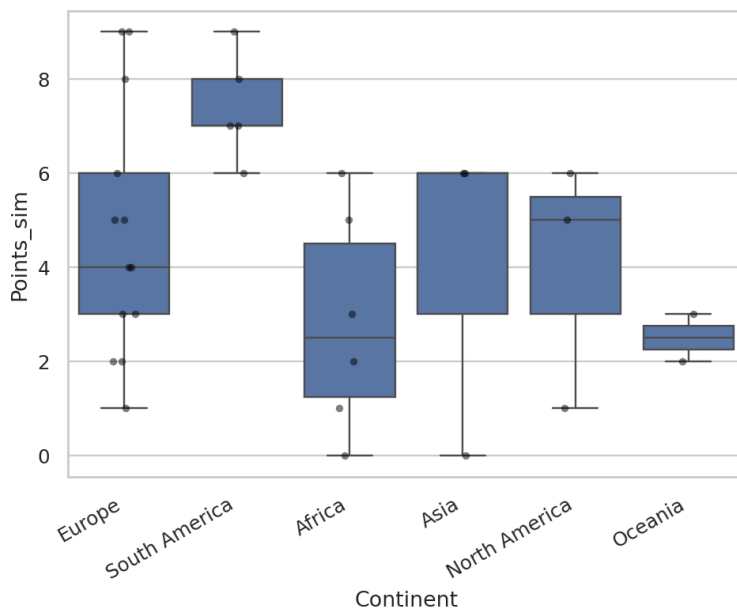
Why a map: performance statistics feel abstract until they are grounded in geography. Plotting each capital turns a spreadsheet into a picture of where footballing power was concentrated.

European capitals form a dense, high-scoring cluster in the north, South American teams post large bubbles despite fewer entries, and Oceania sits geographically and competitively on the periphery — the map tells that story at a glance in a way the bar chart, built on the same data, cannot.

KEY INSIGHT

Footballing power clusters geographically in Europe and South America.

Simulated Points Distribution by Continent



DISTRIBUTION

Simulated Points by Continent

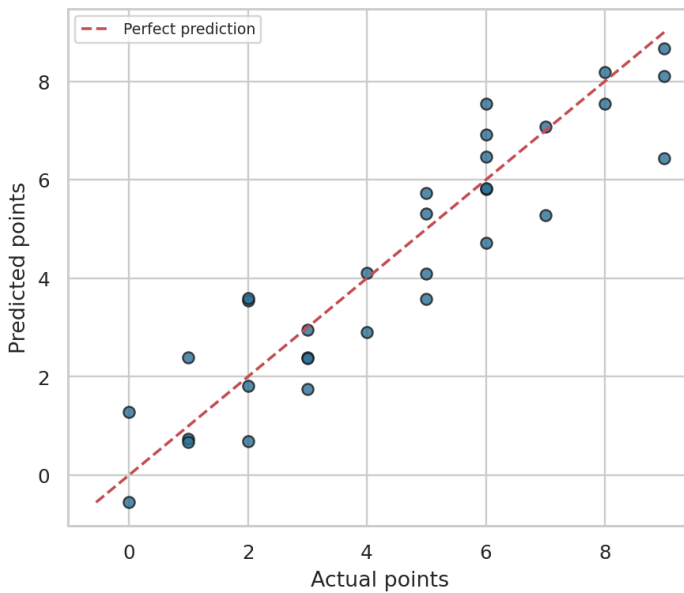
Why a boxplot: comparing six groups on a single metric needs more than an average — a boxplot shows the median, the spread, and outliers side by side, which a bar chart of means would hide.

Europe's median sits highest but its box is also among the widest — a mix of powerhouses and early exits. South America is smaller in number but tightly banded near the top, suggesting more consistent quality across fewer, stronger entries.

KEY INSIGHT

Europe leads on median but South America is more consistent.

Linear Regression: Actual vs. Predicted Points



MACHINE LEARNING - REGRESSION

Actual vs. Predicted Points

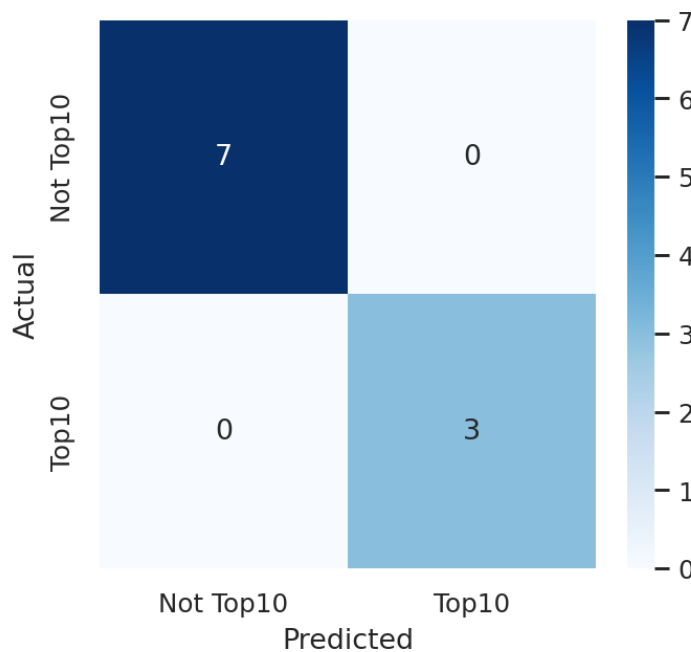
Why a parity plot: for regression, plotting actual against predicted values (with a diagonal reference line) is the standard way to audit fit quality visually, more intuitive than reading R-squared alone.

Most points sit close to the dashed diagonal, matching the model's solid R-squared on goal difference and possession alone. The few visible outliers are a reminder that a two-feature linear model will never fully capture a noisy, simulated target.

KEY INSIGHT

Goal difference and possession explain most of the variance in points.

Perceptron Confusion Matrix



MACHINE LEARNING - CLASSIFICATION

Perceptron Confusion Matrix

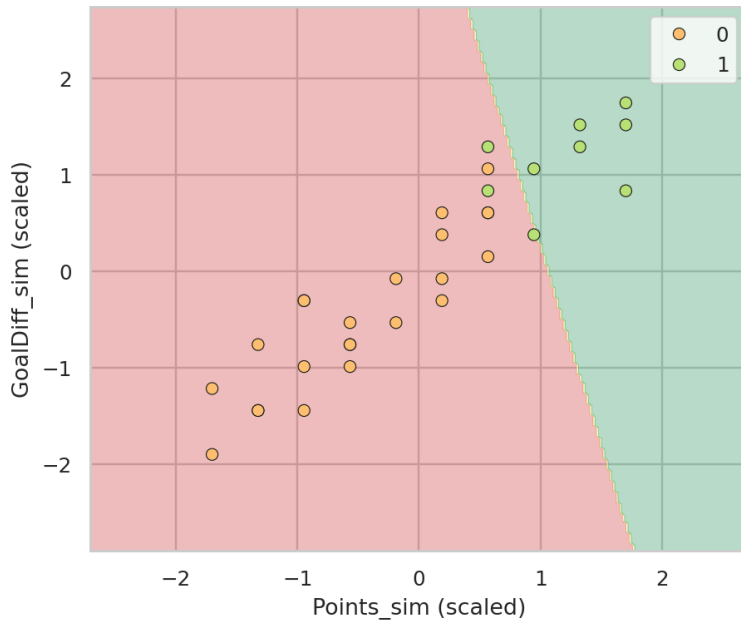
*Why a confusion matrix: for a yes/no prediction, accuracy alone hides *how* a model is wrong. The matrix separates false positives from false negatives, which matter differently in practice.*

The Perceptron cleanly separates Top-10 finishers from the rest on this small test split, with errors concentrated in the 'nearly there' zone — teams whose stats sit right on the boundary rather than clearly on one side.

KEY INSIGHT

The model separates Top-10 teams cleanly, erring only near the cutoff.

Perceptron Decision Boundary



MACHINE LEARNING - CLASSIFICATION

Perceptron Decision Boundary

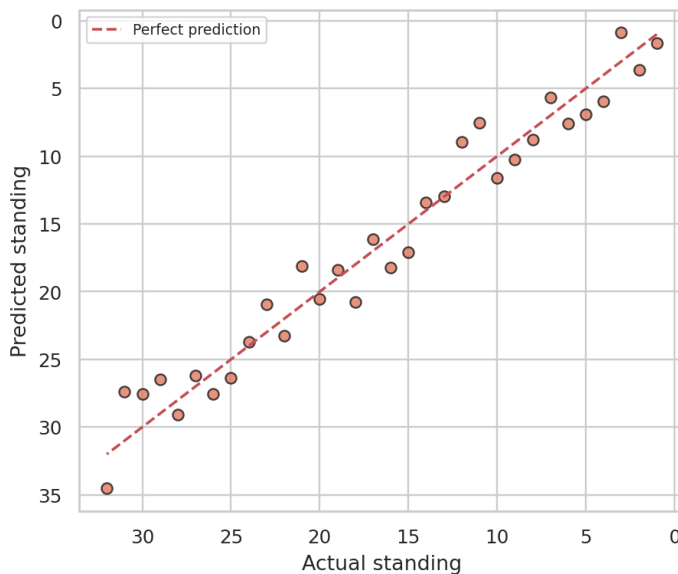
Why a decision-boundary plot: it turns an abstract linear classifier into something visible — literally the line the model draws through feature space — which builds far more intuition than a coefficient table.

Top-10 teams occupy a coherent region of high points and strong goal difference, and the Perceptron's boundary tracks that region closely — visual confirmation that the two chosen features carry real separating power, not just noise.

KEY INSIGHT

Two features alone carry real, visible separating power.

Prediction Model: Actual vs. Predicted Standing



MACHINE LEARNING - PREDICTION

Predicted vs. Actual Standing

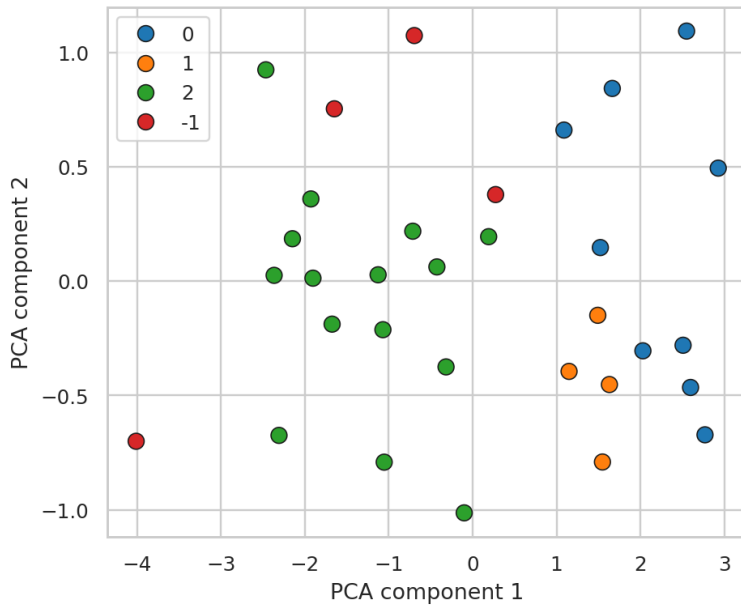
Why the same parity-plot grammar again: reusing the actual-vs-predicted layout from the regression page lets a reader compare 'points' prediction quality against 'standing' prediction quality without learning a new chart type.

Axes are inverted so better (lower-numbered) standings sit toward the top-right, matching intuition. The trend is directionally right but noisier than the points model — a fair outcome, since standing depends on far more than five match statistics.

KEY INSIGHT

Standing is harder to predict than points -- it depends on more than stats.

DBSCAN Clusters (PCA-reduced)



MACHINE LEARNING - CLUSTERING

DBSCAN Clusters (PCA-Reduced)

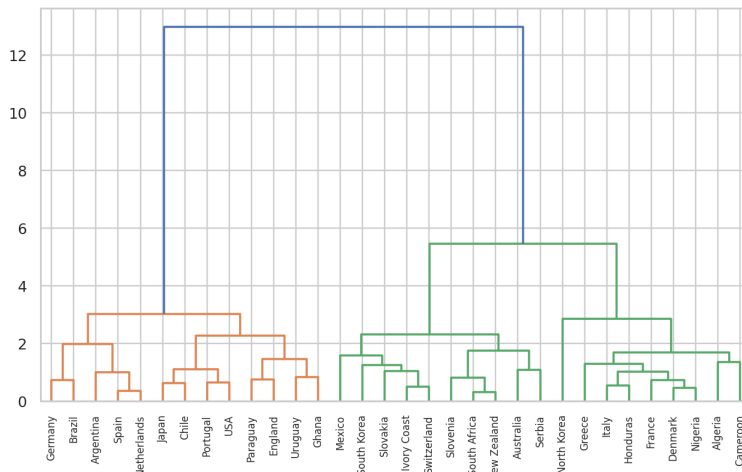
Why PCA plus scatter: DBSCAN clusters on four dimensions at once, which can't be plotted directly. Projecting onto the two principal components that capture the most variance makes the grouping visible without misrepresenting it.

DBSCAN recovers two to three density-based tiers of performance plus a handful of noise points — teams whose stat profile does not fit neatly into either the elite or the mid-table group, which is exactly the kind of edge case density clustering is designed to flag.

KEY INSIGHT

Density clustering surfaces 2-3 tiers plus a handful of edge cases.

Hierarchical Clustering Dendrogram (Ward linkage)



MACHINE LEARNING - CLUSTERING

Ward-Linkage Dendrogram

*Why a dendrogram: unlike DBSCAN's flat output, a dendrogram shows the *entire* merge history, so a reader can choose their own number of clusters after the fact instead of trusting one fixed cutoff.*

The tree splits first into two broad performance tiers, each of which then branches into tighter sub-groups of similar teams — structure that echoes the DBSCAN result but with the added nuance of **how** closely each team resembles its neighbors.

KEY INSIGHT

The full merge tree confirms the tiers DBSCAN found, with finer nuance.